



NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

PYTHON FOR DATA ANALYTICS

PROJECT SUBMISSION

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TITLE: HR Employee Attrition Analysis

GITHUB LINK : <https://github.com/sugumaranj/HR-Employee-Attrition-Analysis.git>

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1. ABSTRACT

Project Overview

This project analyzes a comprehensive Human Resources dataset containing 1,470 employee records to identify the core factors driving employee turnover. High attrition rates cost companies significant time and money, making it crucial to understand exactly why employees choose to leave.

To investigate this, the project utilized Python and the Pandas library to clean the raw data, removing redundant columns like employee counts and standard hours, reducing the dataset from 35 to 31 purely analytical features. From there, Exploratory Data Analysis (EDA) was conducted using Pandas grouping functions to calculate attrition percentages across various employee demographics and job roles. Matplotlib was then used to build clear visual



representations of these findings.

The analysis revealed an overall baseline attrition rate of 16.1%. Key insights showed that employees working overtime, those in the Sales department (specifically Sales Representatives at nearly 40% attrition), and frequent business travelers are at the highest risk of leaving. Furthermore, the data clearly demonstrated that lower average monthly income (around \$4,800 for those leaving versus \$6,800 for those staying) and lower job satisfaction heavily impact turnover. The final outcome of this project is a data-driven overview that HR departments can use to implement targeted retention strategies, specifically focusing on workload balance and compensation structures for high-risk roles.

2. PROBLEM STATEMENT

Business Problem / Real-World Challenge

The primary challenge this project addresses is high employee turnover, commonly known as employee attrition. When a company constantly loses its employees, it faces significant financial costs related to recruiting, hiring, and training new staff. Beyond the monetary cost, high attrition disrupts team workflows, lowers overall workplace morale, and delays project timelines.

Employees choose to leave for a variety of reasons, such as inadequate compensation, burnout from excessive overtime, poor work-life balance, or dissatisfaction with their specific job roles. This issue directly affects HR managers who struggle to retain talent, department heads who lose experienced team members, and the executive leadership whose bottom line is impacted by constant hiring costs.

Data analysis is essential to solve this problem because HR departments cannot rely on guesswork or exit interviews alone to understand why people quit. By systematically analyzing the company's dataset, we can identify hidden patterns and mathematical correlations that highlight exactly who is leaving and why.

The expected improvement from this analysis is that the company will stop using a "one-size-fits-all" approach to employee retention. Instead, management will be able to make targeted, data-driven decisions, such as restructuring the workload for high-risk roles like Sales Representatives, adjusting pay scales for lower-income brackets, or limiting mandatory overtime, to actively prevent valuable employees from leaving.

3. OBJECTIVES

The primary objective of this project focuses on generating meaningful insights from the HR employee dataset to identify the root causes of staff turnover. By systematically analyzing the relationships between employee demographics, working conditions, and attrition rates, the project aims to improve the overall understanding of data-driven decision-making processes within human resources management, moving away from intuition-based retention strategies toward evidence-based solutions.

4. PROJECT SCOPE

This project focuses on analyzing and visualizing workforce data using Python in Jupyter Notebook to generate meaningful insights from datasets. The analytical scope is strictly confined to the provided dataset of 1,470 employees. The workflow encompasses data ingestion of the raw CSV file, systematic data cleaning to remove unneeded variables, exploratory data analysis (EDA) to calculate attrition percentages across key metrics (such as department, salary, experience, and work-life balance), and the creation of targeted visual graphs to effectively communicate these trends to stakeholders.

5. TECHNOLOGIES USED

Development Environment

- **Programming language:** Python 3.13.9
- **Environment:** Jupyter Notebook

Libraries Used and Dataset Source

- **Pandas (v2.3.3):** This library served as the backbone for data handling and manipulation. It was used extensively for reading the dataset, cleaning the data (dropping non-variable columns), and performing Exploratory Data Analysis. The Pandas GroupBy function was heavily utilized to aggregate attrition counts and calculate accurate percentages across different employee subgroups.
- **NumPy (v2.3.5):** Used alongside Pandas to handle numerical operations, array

transformations, and mathematical calculations required during the data preprocessing stages.

- **Matplotlib (v3.10.6):** The primary visualization library used to translate the raw analytical findings into easily understandable visual formats. It was used to generate the pie charts and bar graphs that highlight the attrition trends.
- **Dataset Source:** The dataset used in this project is the IBM HR Analytics Employee Attrition & Performance dataset. It contains 1,470 employee records with demographic, job-related, compensation, and satisfaction attributes used for attrition analysis.

Data Source: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

6. APPLICATION FLOW

Step-by-Step Workflow

The application workflow is systematically divided into four distinct Jupyter Notebooks to maintain a clean and modular data analysis pipeline:

1. **Data Understanding** (01_Data_Understanding.ipynb): The process begins by loading the raw `employee_attrition.csv` dataset into a Pandas DataFrame to inspect its initial structure, check the total number of records (1,470 rows), and understand the variables present.
2. **Data Cleaning** (02_Data_Cleaning.ipynb): The dataset is preprocessed by identifying and dropping unnecessary or zero-variance columns (reducing the dataset from 35 to 31 columns). The resulting refined data is exported and saved as `employee_cleaned.csv` within the `data/CleanedDataset/` directory.
3. **Exploratory Data Analysis** (03_EDA_and_Insights.ipynb): This phase involves deep analysis of the cleaned data. The workflow uses Pandas grouping techniques to segment the workforce and calculate exact attrition percentages across various factors. Aggregated statistical summaries, such as `department_attrition.csv`, are exported to the `results/` folder.

4. **Data Visualization** (04_Visualizations.ipynb): In the final step, the analyzed metrics are processed using Matplotlib to generate visual graphs. The workflow produces 10 specific charts highlighting attrition trends, which are saved directly into the charts/ directory.

7. KEY PYTHON DATA ANALYTICS CONCEPTS USED

1. Data Handling

- Utilizing pandas to read and manipulate data frames (`pd.read_csv()`).
- Managing directory paths to successfully load raw data and export cleaned datasets and result tables to specific project folders.

2. Data Cleaning

- Implementing dimensionality reduction by dropping specific columns that hold no analytical value, ensuring the machine processes only relevant business features.

3. Exploratory Data Analysis (EDA)

- Applying the Pandas `groupby()` function to aggregate workforce data.
- Calculating descriptive statistics to find the proportion of employees leaving within specific categories (e.g., measuring attrition by Job Role, Income Level, and Years of Experience).

4. Visualization

- Creating pie charts with percentage labels to display the overall attrition distribution.
- Designing horizontal and vertical bar graphs using Matplotlib to visually compare attrition rates across different departments and compensation levels.

8. EXPECTED OUTPUT

Final Results

Dataset & Results Previews:

The screenshot displays a JupyterLab environment with a notebook titled "01_Data_Understanding". The notebook is open to a code cell labeled "[18]: # Dataset structure". The code executed is "df.info()", which outputs the following information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
 #   Column              Non-Null Count  Dtype  ---
 0   Age                 1470 non-null   int64  
 1   Attrition            1470 non-null   object 
 2   BusinessTravel       1470 non-null   object 
 3   DailyRate            1470 non-null   int64  
 4   Department           1470 non-null   object 
 5   DistanceFromHome     1470 non-null   int64  
 6   Education            1470 non-null   int64  
 7   EducationField       1470 non-null   object 
 8   EmployeeCount        1470 non-null   int64  
 9   EmployeeNumber       1470 non-null   int64  
10  EnvironmentSatisfaction 1470 non-null   int64  
11  Gender              1470 non-null   object 
12  HourlyRate           1470 non-null   int64  
13  JobInvolvement       1470 non-null   int64  
14  JobLevel             1470 non-null   int64  
15  JobRole              1470 non-null   object 
16  JobSatisfaction      1470 non-null   int64  
17  MaritalStatus        1470 non-null   object 
18  MonthlyIncome         1470 non-null   int64  
19  MonthlyRate          1470 non-null   int64  
20  NumCompaniesWorked    1470 non-null   int64  
21  Over18               1470 non-null   object 
22  OverTime             1470 non-null   object 
23  PercentSalaryHike     1470 non-null   int64  
24  PerformanceRating     1470 non-null   int64  
25  RelationshipSatisfaction 1470 non-null   int64  
26  StandardHours        1470 non-null   int64  
27  StockOptionLevel     1470 non-null   int64  
28  TotalWorkingYears     1470 non-null   int64  
29  TrainingTimesLastYear 1470 non-null   int64  
30  WorkLifeBalance      1470 non-null   int64  
31  YearsAtCompany        1470 non-null   int64  
32  YearsInCurrentRole    1470 non-null   int64  
33  YearsSinceLastPromotion 1470 non-null   int64  
34  YearsWithCurrManager  1470 non-null   int64  
dtypes: int64(26), object(9)
```



Cleaned Data Output:

```
HR-Employee-Attrition-Analysis/n... Home
02_Data_Cleaning
01_Data_Understanding
localhost:8888/notebooks/Desktop/NM%20Internship/FinalProject/HR-Attrition-Analysis/notebooks/02_Data_Cleaning.ipynb
jupyter 02_Data_Cleaning Last Checkpoint: 3 days ago
File Edit View Run Kernel Settings Help Trusted
dtypes: int64(23), object(8)
memory usage: 356.1+ KB

[16]: cleaned_df.head()

[16]:
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	...	PerformanceRating
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	2	Female	...	3
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	3	Male	...	4
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	4	Male	...	3
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	4	Female	...	3
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	Male	...	3

5 rows x 13 columns

```
cleaned_df.tail(10)

[39]:
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EnvironmentSatisfaction	Gender	...	PerformanceRating
1460	29	No	Travel_Rarely	460	Research & Development	28	4	Medical	4	Female	...	...
1461	50	Yes	Travel_Rarely	410	Sales	28	3	Marketing	4	Male	...	...
1462	39	No	Travel_Rarely	722	Sales	24	1	Marketing	2	Female	...	...
1463	31	No	Non-Travel	325	Research & Development	5	3	Medical	2	Male	...	...
1464	26	No	Travel_Rarely	1167	Sales	5	3	Other	4	Female	...	...
1465	36	No	Travel_Frequently	884	Research & Development	23	2	Medical	3	Male	...	...
1466	39	No	Travel_Rarely	613	Research & Development	6	1	Medical	4	Male	...	...
1467	27	No	Travel_Rarely	155	Research & Development	4	3	Life Sciences	2	Male	...	...
1468	49	No	Travel_Frequently	1023	Sales	2	3	Medical	4	Male	...	...
1469	34	No	Travel_Rarely	638	Research &	8	3	Medical	2	Male	...	...

[illegible]

HR-Employee-Attrition-Analysis | Home | department_attrition.csv | 02_Data_Cleaning | 01_Data_Understanding | + | - | Chat

localhost:8888/edit/Desktop/NM%20Internship/FinalProject/HR-Attrition-Analysis/results/department_attrition.csv

Jupyter department_attrition.csv Last Checkpoint: 2 days ago

File Edit View Settings Help

Delimiter: ,

	Department	No	Yes
1	Human Resources	80.95	19.05
2	Research & Development	86.16	13.84
3	Sales	79.37	20.63

sugumaran/HR-Employ | Home | 03_EDA_and_Insights | department_attrition.csv | 02_Data_Cleaning | 01_Data_Understanding | + | - | Chat

localhost:8888/notebooks/Desktop/NM%20Internship/FinalProject/HR-Attrition-Analysis/notebooks/03_EDA_and_Insights.ipynb

Jupyter 03_EDA_and_Insights Last Checkpoint: 3 days ago

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Trusted

JupyterLab Python 3 (pykernel)

EDA Summary & Conclusion

Key Insights

- Overall Attrition:** The company has an employee attrition rate of **16.12%**.
- Highest Risk Department:** The **Sales Department** shows the highest attrition rate (20.63%) among all departments.
- Highest Risk Role:** **Sales Representatives** exhibit the highest attrition rate (39.76%) among all job roles.
- Burnout Factors:** Employees working **Overtime** have an attrition rate of 30.53%, while employees who travel **Frequently** for business show an attrition rate of 24.91%.
- Employee Well-being:** Lower **Work-Life Balance** and **Job Satisfaction** levels are associated with higher attrition rates.
- Compensation & Experience:** Employees who left earned an average monthly income of **₹4,787.09**, compared to **₹6,832.74** for employees who stayed.
- Experience & Tenure:** Employees who left had an average of **8.24 years** of work experience and **5.13 years** at the company, compared to **11.86 years** and **7.37 years** respectively for retained employees.

Conclusion & Recommendations

The analysis indicates that employee attrition is strongly associated with overtime work, frequent business travel, lower satisfaction levels, lower compensation, and shorter tenure. Sales-related roles, particularly Sales Representatives, exhibit the highest turnover rates.

Recommendations

- Prioritize Sales Retention:** Develop targeted retention programs for Sales Representatives and employees in the Sales department.
- Reduce Burnout:** Monitor overtime workloads and explore alternatives to frequent business travel where possible.
- Improve Employee Well-being:** Strengthen work-life balance initiatives and employee satisfaction programs.
- Support Early-Career Employees:** Review compensation structures, onboarding processes, and career development opportunities for less experienced employees.

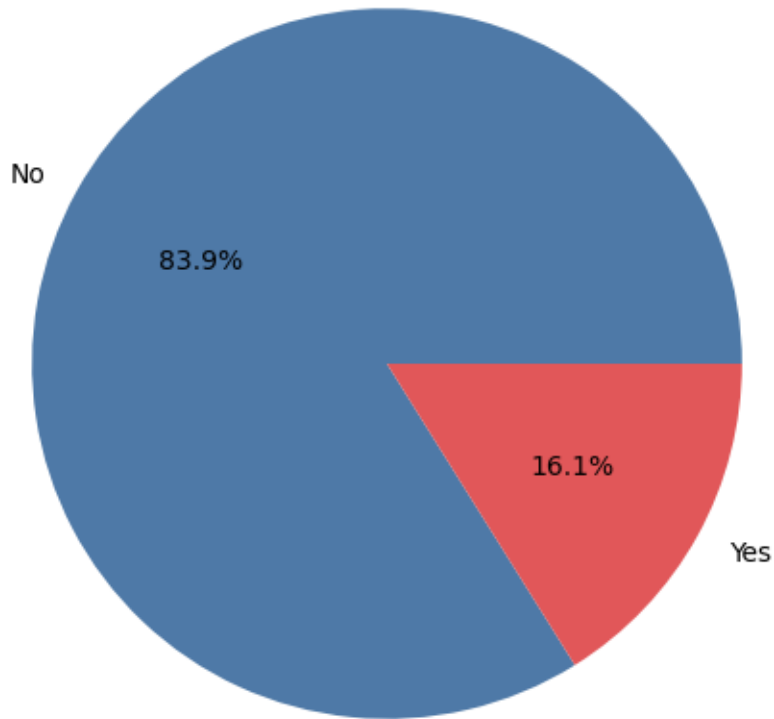
Correlation Analysis

```
[25]: df["AttritionNumeric"] = df["Attrition"].map({
      "No": 0,
      "Yes": 1
    })
```

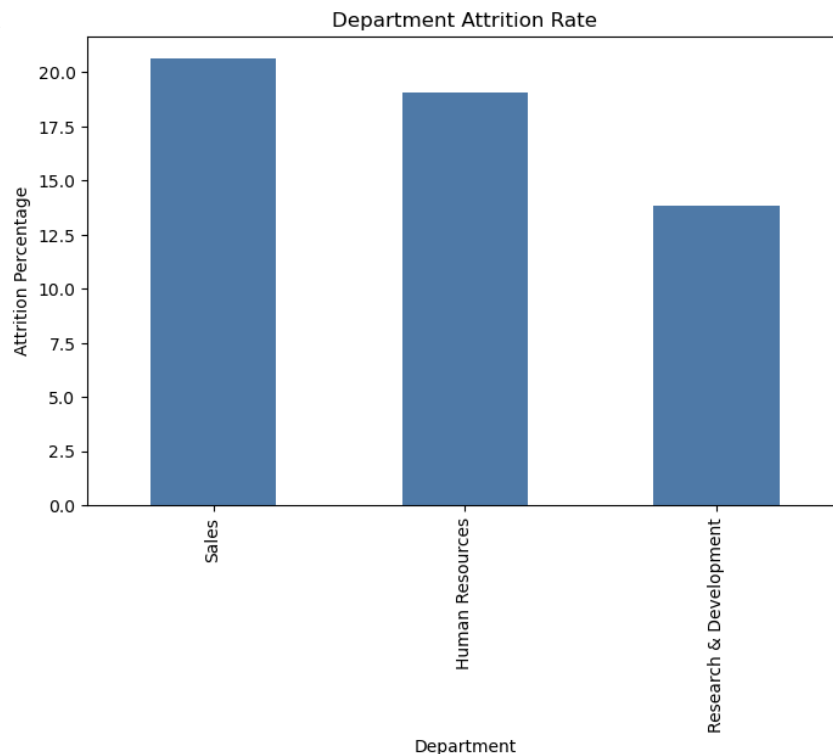
Visualizations Created:

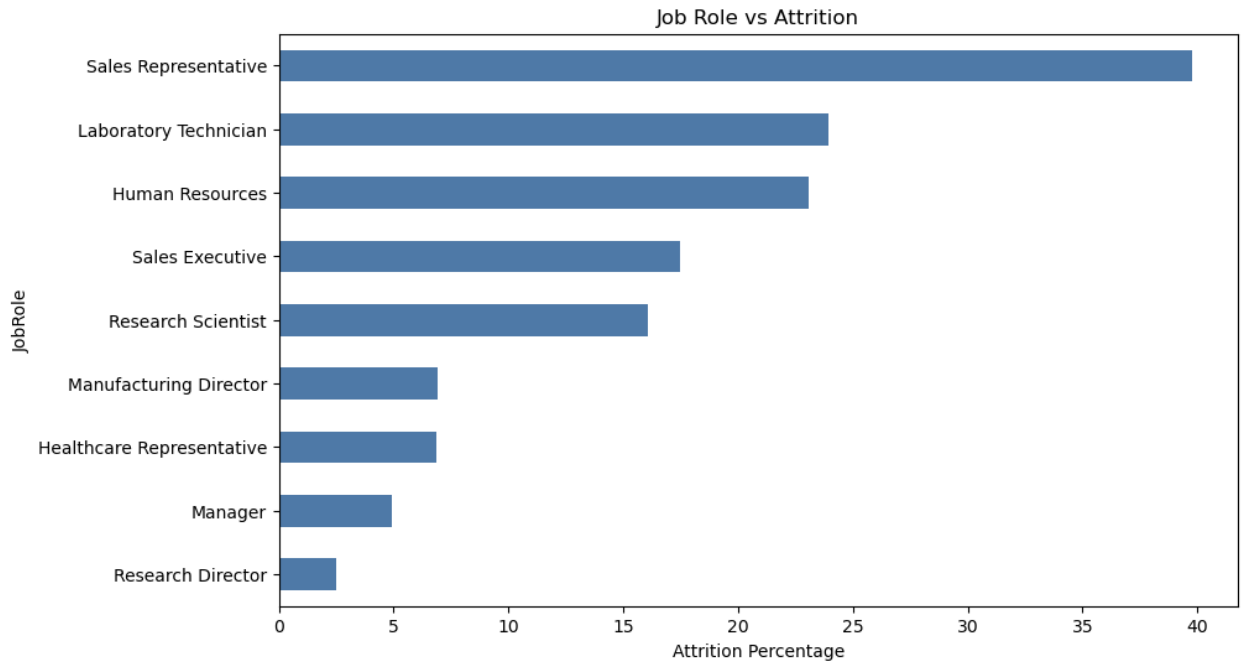
Overall Attrition: A pie chart (charts/attrition_distribution.png) showing the 16.1% baseline turnover rate.

Employee Attrition Distribution

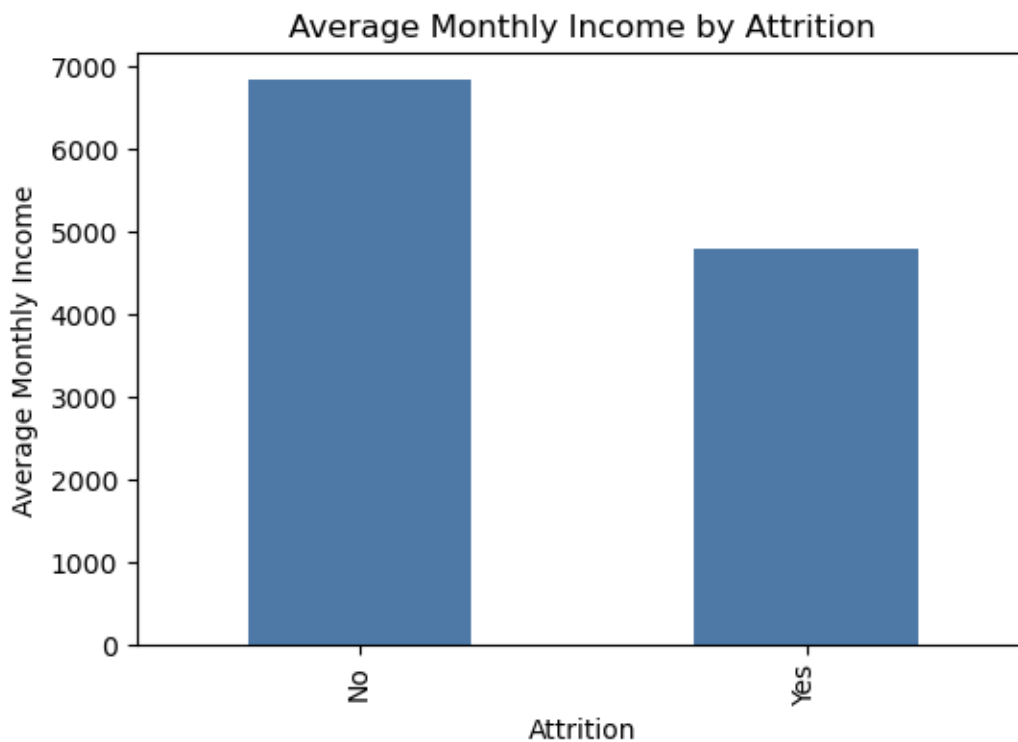


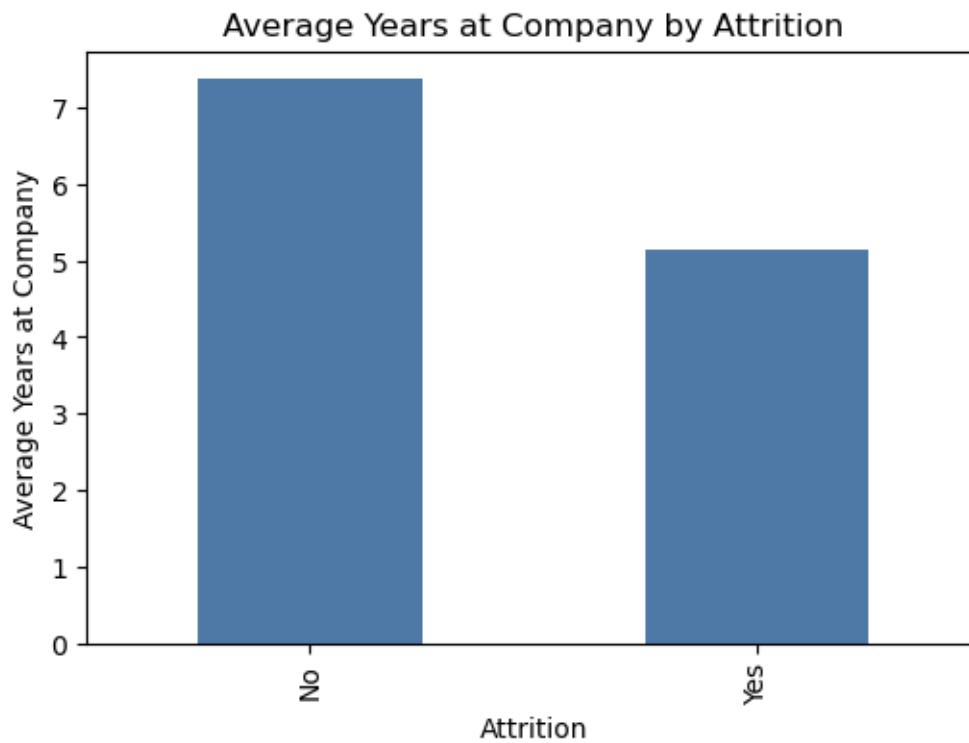
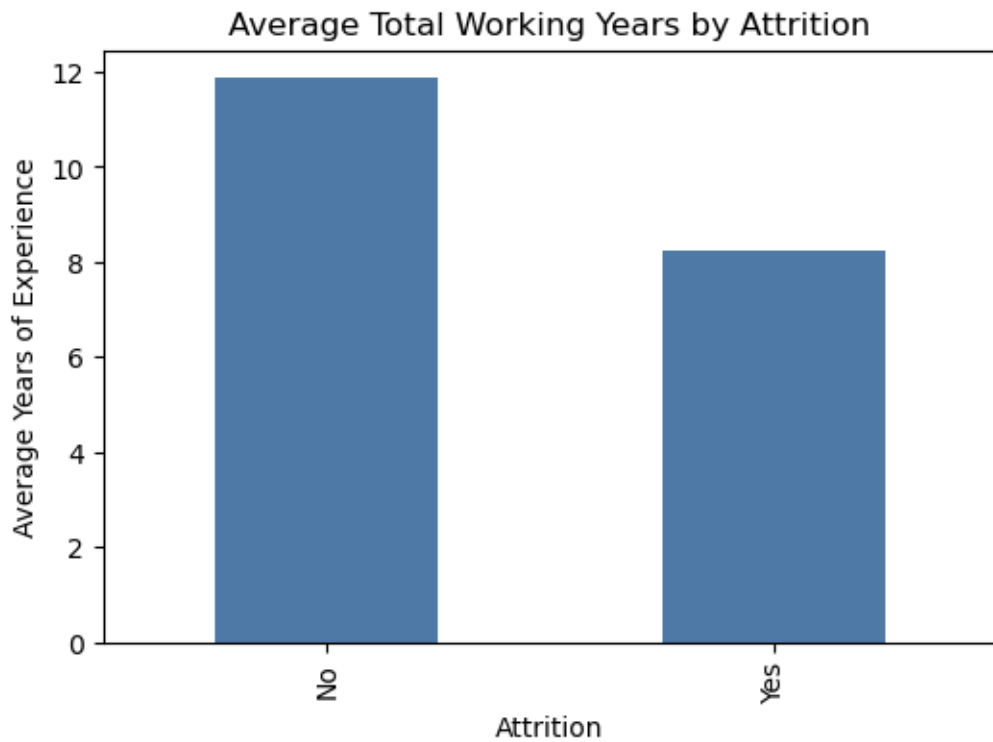
Department & Role Trends: Bar charts (charts/department_attrition.png and charts/jobrole_attrition.png) revealing Sales Representatives have the highest attrition at nearly 40%.





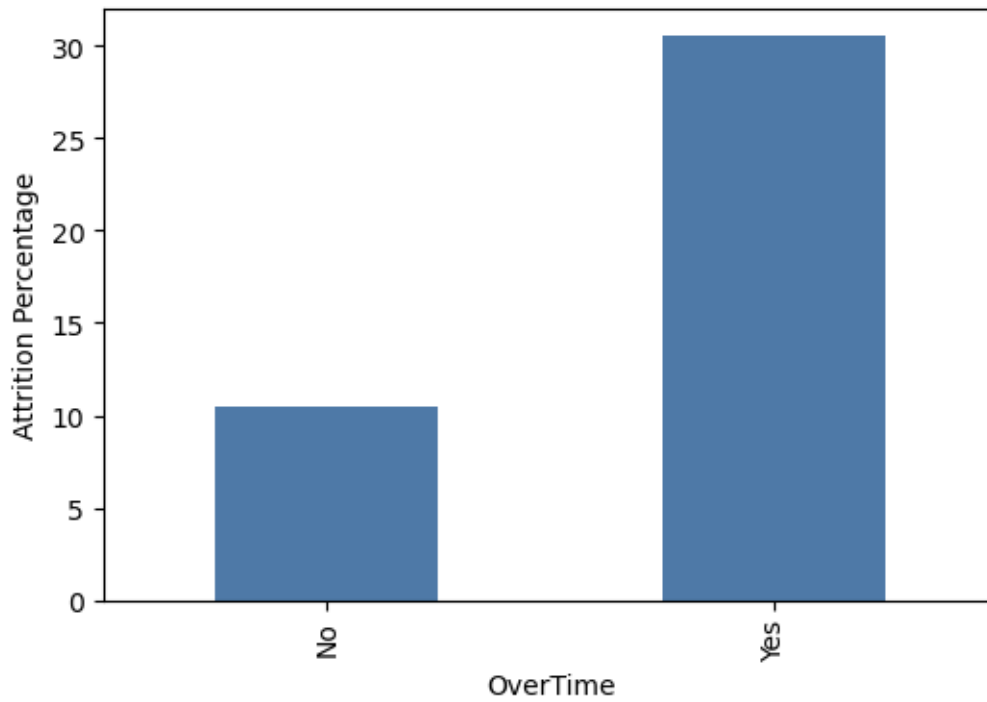
Financial & Experience Impact: Graphs (charts/monthly_income_attrition.png, charts/experience_attrition.png, and charts/tenure_attrition.png) demonstrating that employees who leave have lower average salaries, fewer total working years, and shorter tenures at the company.



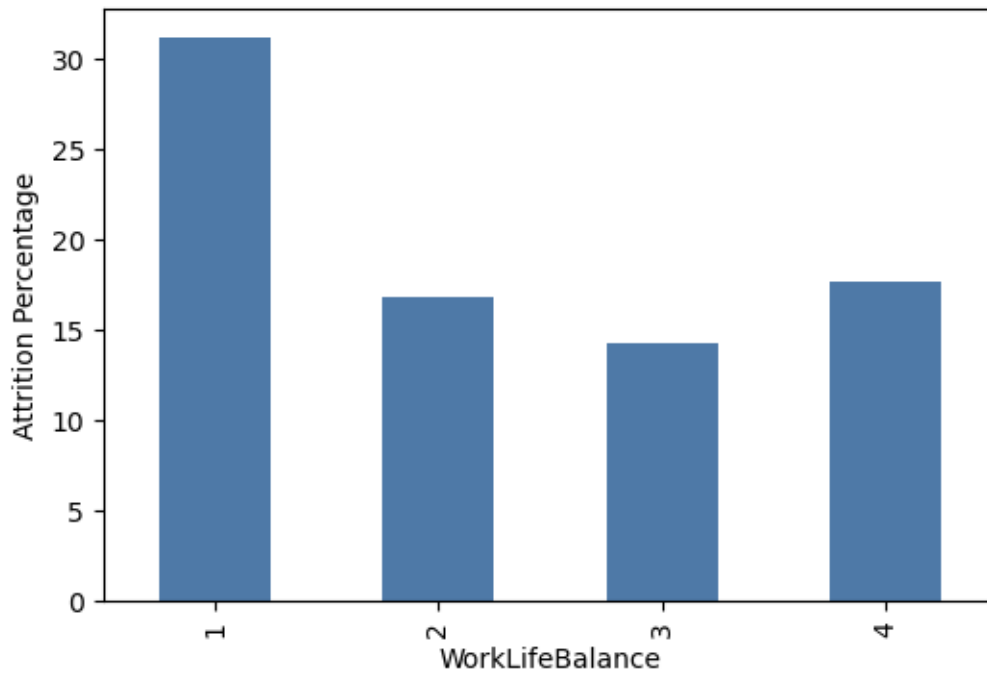


Work Conditions: Charts (charts/overtime_attrition.png, charts/worklifebalance_attrition.png, charts/business_travel_attrition.png, and charts/job_satisfaction_attrition.png) illustrating that excessive overtime, low work-life balance, high travel demands, and low job satisfaction strongly correlate with employees quitting.

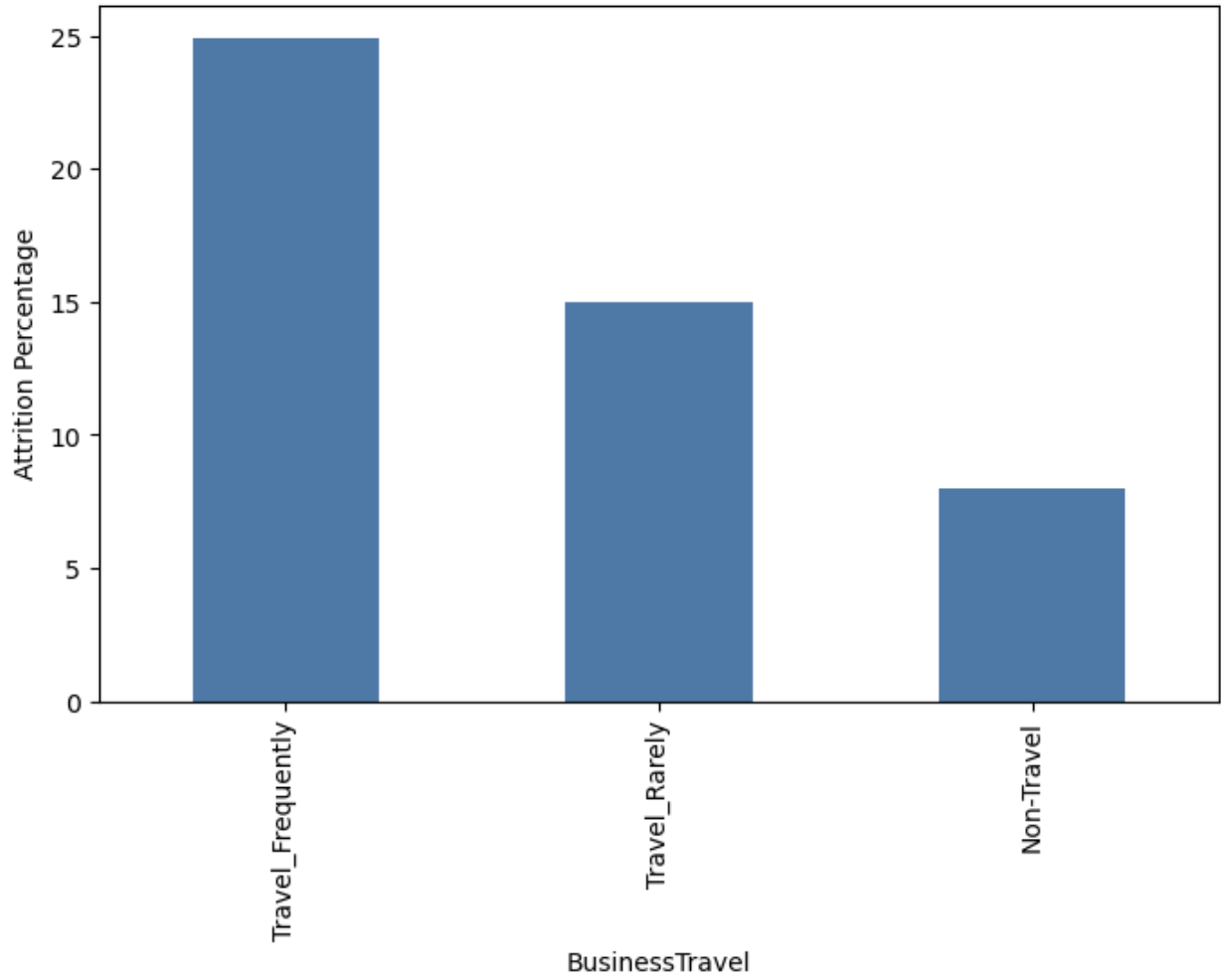
Overtime vs Attrition



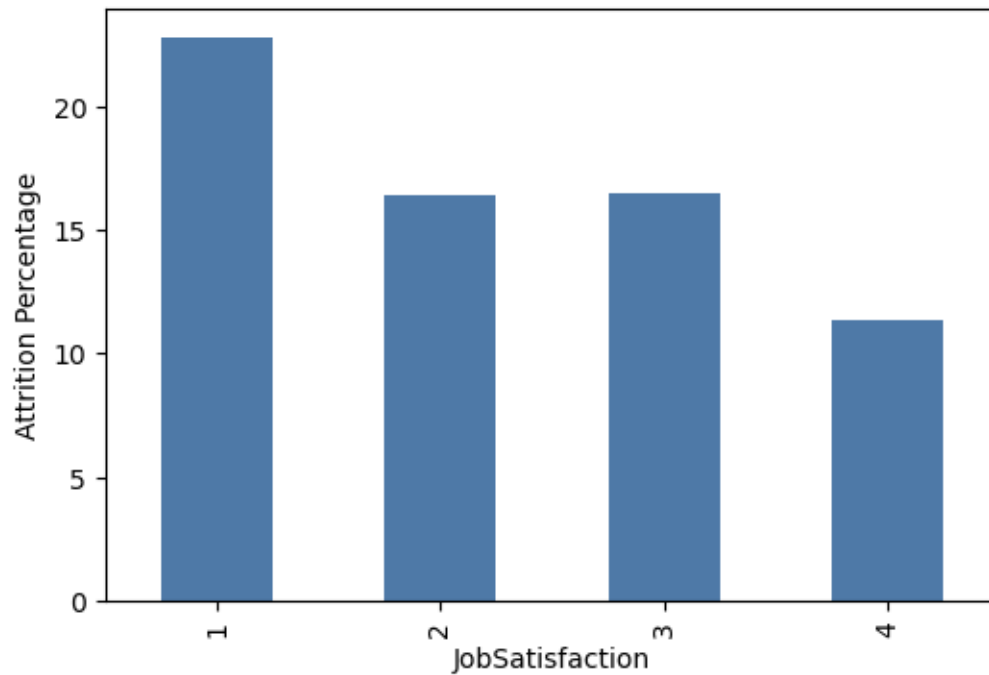
Work-Life Balance vs Attrition



Business Travel vs Attrition



Job Satisfaction vs Attrition



The project screenshots includes dataset preview, cleaned data after preprocessing, and visualizations created using Python in Jupyter notebook.

9. CHALLENGES FACED

Difficulties Encountered

During the development of this project, several challenges were encountered:

Understanding the Dataset Structure

Initially, it was difficult to identify which columns were useful for analysis and which columns had no analytical value. This challenge was addressed by carefully exploring the dataset using Pandas functions such as `head()`, `info()`, `describe()`, and `nunique()`. As a result, unnecessary columns like `EmployeeCount`, `Over18`, `StandardHours`, and `EmployeeNumber` were identified and removed.

Learning Pandas GroupBy Operations

Since I was new to data analytics with python, understanding how to use the `groupby()` function to analyze attrition across departments, job roles, overtime, and other factors was initially challenging. Through practice given by mentor and experimentation, I learned how to aggregate data, calculate attrition counts, and compute attrition percentages accurately.

Calculating Attrition Percentages Correctly

Converting raw employee counts into meaningful attrition percentages was another challenge. To solve this, percentage calculations were implemented using Pandas operations, allowing fair comparisons across different categories and employee groups.

Selecting Meaningful Visualizations

Choosing appropriate chart types to represent business insights was difficult at first. Different visualization approaches were evaluated, and bar charts and pie charts were selected because they clearly communicated attrition patterns and trends.

Interpreting Business Insights from Data

Transforming numerical results into actionable business conclusions required careful analysis. This challenge was overcome by comparing attrition rates across departments, job roles, salary levels, work-life balance scores, and overtime categories to identify the most important factors associated with employee attrition.

Through these challenges, I gained practical experience in data cleaning, exploratory data analysis, data visualization, and business insight generation using Python and Pandas.

10. LEARNING OUTCOMES

Skills Gained

Technical Learning:

- **Python for Data Science:** Gained deep practical experience using Python as an analytical tool, moving beyond basic programming into data manipulation.
- **Pandas Mastery:** Learned how to efficiently load external CSV files, clean missing or irrelevant data, and perform complex aggregations using grouping functions.
- **Data Visualization:** Mastered the use of Matplotlib to create diverse chart types (pie charts and bar graphs etc...) and learned how to format them for professional presentations, and also learned to create a scatter plot and heat map in seaborn and I implemented in mini project of the internship
- **Environment Management:** Gained experience managing a multi-stage project workflow across multiple Jupyter Notebooks.

Practical Experience:

- **Business Intelligence:** Learned how to look at a massive grid of numbers and extract a cohesive, real-world story. I now understand how data directly informs business strategy.
 - **Problem-Solving:** Developed the ability to approach an open-ended business problem (like "Why are people leaving?") and systematically break it down into testable, data-driven hypotheses.
-

11. CONCLUSION

The HR Employee Attrition Analysis project was highly successful in fulfilling its core objective: transforming raw workforce data into clear, actionable business insights. By systematically cleaning the dataset and applying targeted exploratory data analysis using Python, the project effectively solved the problem of HR departments relying on intuition to understand staff turnover.

Through the implementation of Pandas and Matplotlib, the analysis revealed a 16.1% baseline attrition rate and successfully isolated the primary risk factors. The visual evidence clearly demonstrated that excessive overtime, lower comparative salaries, and specific high-stress positions, most notably the Sales Representative role are the major factors associated with attrition of employees leaving the company.



Ultimately, this project achieved a complete, data-backed dashboard that company leadership can now use to immediately restructure workloads, adjust compensation for vulnerable roles, and implement targeted retention strategies, thereby saving the organization significant time and financial resources.